	AzureSky		
	Statistical Design	Document no: ASMF 7.2.2	
	Compiled by:	**Tshepo Maponya – Laboratory Supervisor**	
	Approved by:	**Sifiso Nyambi – Laboratory Manager**	

Procedure for Statistical Design

This procedure outlines the statistical design for the coal proficiency testing (PT) scheme. It provides detailed specifications in accordance with ISO/IEC 17043 and the statistical methods described in ISO 13528:2022. This document is designed to satisfy SANAS F91 clause 7.2.2 requirements and to demonstrate that Azuresky's PT scheme is statistically valid, traceable, and fit for purposes.

1. Design Elements and Procedure Specifications

PT Scheme / Measurands	Analysis of Coal: Inherent Moisture (SANS 5925), Volatile Matter (ISO 562), Calorific Value (ISO 1928), Ash (ISO 1171), Total Sulphur (ASTM D4239), Ash Fusion Temperature, Hardgrove Grindability Index (HGI), Ash in Analysis (AI).
PT Item & Frequency (7.2.2.3d, 7.2.2.3j)	Item: One bottle (approx. 150 g) of pulverized coal. Frequency: Monthly PT rounds.
Objectives (7.2.2.1)	1. To evaluate the performance of participating laboratories through interlaboratory comparison. 2. To establish robust consensus values for each measurand to assign property values. 3. To promote continuous improvement in coal analysis laboratories.
Data Type (7.2.2.1)	Quantitative – all measurands are reported as numerical results.
Statistical Assumptions & Error Types (7.2.2.1)	It is assumed that participant results are independent and approximately normally distributed after application of robust statistical treatment. Both random and systematic errors may occur among laboratories; the robust method (Algorithm

	A) minimizes the impact of such variations.
Determination of Assigned Value (xA) (7.2.2.3a)	The assigned value is the robust consensus mean (xA) calculated using ISO 13528:2022 Algorithm A. This approach is chosen for its robustness and resistance to the influence of extreme values.
Determination of σpt (7.2.2.3e)	The standard deviation for proficiency assessment (σpt) is determined as the robust consensus standard deviation per ISO 13528:2022 Algorithm A.
Uncertainty of Assigned Value (7.2.2.3a)	The uncertainty associated with the assigned value is estimated as $u(xA) = \sigma pt / \sqrt{n}$, where n is the number of valid participant results.
Performance Evaluation (7.2.2.2)	Participant performance is assessed using z-scores: $z = (xi - xA) / \sigma pt$. Interpretation: $\|z\| \leq 2.0$ – Acceptable $2.0 < \|z\| < 3.0$ – Questionable $\|z\| \geq 3.0$ – Unacceptable.
Minimum Participants (n) (7.2.2.3b)	A minimum of 10 participants is required for statistically meaningful evaluation.
Alternative Approach (n < 10) (7.2.2.3b)	If fewer than 10 participants submit results, the assigned value (xA) will be determined by the PT Provider's reference laboratory and the σpt estimated from historical data. This deviation will be recorded and disclosed in the final report.
Handling of Outliers (7.2.2.3h, 7.2.2.3i)	No data are removed before analysis; Algorithm A provides robustness against outliers. However, results identified as $\|z\| \geq 3.0$ are flagged as outliers. Invalid results (e.g., incorrect method, units, or incomplete data) may be excluded and documented with justification.
Handling of Test Methods (7.2.2.3f)	All specified participant methods (SANS, ISO, ASTM) are considered technically equivalent and are combined into a single statistical population.

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Measurement Uncertainty (7.2.2.3g)	Participants are not required to report their measurement uncertainty. Performance evaluation is based solely on z-scores; however, uncertainty of the assigned value is calculated and reported by the PT provider.
Significant Figures & Reporting (7.2.2.3c)	Results are reported to the following decimal places: 1 decimal for Ash, IM, VM; 2 decimals for CV, TS; 0 decimals for HGI, AI.

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2. Narrative Statistical Design Description

The AzureSky coal proficiency testing scheme is designed for quantitative determination of coal characteristics. Each participant receives one bottle of pulverized coal and reports single-determination results for the defined measurands. Statistical design ensures that performance evaluation is objective, consistent, and in line with ISO 13528:2022.

The statistical model is based on robust consensus statistics (Algorithm A), which assume that valid participant results follow an approximately normal distribution after removal of strong bias and the influence of extreme values. This model provides stable estimates of central tendency and dispersion, even in small datasets.

AzureSky verifies the reasonableness of statistical assumptions by reviewing data distributions, z-score patterns, and historical control charts from previous PT rounds. The PT provider ensures that all analyses are performed strictly according to ISO 13528 procedures, and records are maintained for SANAS review.

In cases where insufficient data are available, or when non-equivalent methods are identified, AzureSky will apply alternative approaches and transparently document these adjustments in the PT report to maintain fairness and traceability. This document shall be reviewed every three years or upon revision of ISO 13528 or ISO/IEC 17043.

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3. Revision table

Revision no	Content changed	Reason for revision	Date changed
01	• Add assumptions (normality, robust, distribution and handling) • Random and systematic errors are both expected and addressed via the robust consensus analysis. • Algorithm A inclusion: Used to minimise the effects of outliers • Add short formula or note for uncertainty of assigned values. • Add notes for handling invalid/technically incorrect results. • Control charts confirm validity of distribution.	• The statistical design did not cover all the requirements as per the SANAS F91 form.	11/11/2025

Effective date	Rev status	Next review date
11/2025	00	11/2028